

Installing and Using PostgreSQL Modules



In this article, we will learn how to install and use the PostgreSQL modules *chkpass*, *fuzzystrmatch*, *isn* and *hstore*. Modules add different capabilities to a database, like admin and monitoring tools, new data types, operators, functions and algorithms. Let's look at modules that add new data types and algorithms, which will help us to push some of the application logic to the database.

PostgreSQL has been called the 'most advanced open source database'. I have been using it for the last four years as an RDBMS for *Foodlets.in*, and as a spatial data store at CSTEP (Center for Study of Science, Technology and Policy). PostgreSQL is one piece of software that doesn't fail to impress me every now and then.

Installing the modules



Note: I am running Ubuntu 10.04 and PostgreSQL 8.4.

Install the *postgresql-contrib* package and restart the database server, then check the *contrib* directory for the list of available modules:

```
sudo apt-get install postgresql-contrib
sudo /etc/init.d/postgresql-8.4 restart
cd /usr/share/postgresql/8.4/contrib/
ls
```

Create a test database called *module_test*:

```
su postgres
createdb module_test
```

Apply the *chkpass*, *fuzzystrmatch*, *isn* and *hstore* modules to the *module_test* database by running the following commands:

```
psql -d module_test -f chkpass.sql
psql -d module_test -f fuzzystrmatch.sql
psql -d module_test -f isn.sql
psql -d module_test -f hstore.sql
```

Let us now look at an example of how each of the modules is used.

Using *chkpass*

The *chkpass* module will introduce a new data type, 'chkpass', in the database. This type is used to store an encrypted field, e.g., a password. Let's see how *chkpass* works for a user account table that we create and insert two rows into: